writes. In a world without heat, everything would rebound elastically, leaving no trace.[100]

It is the presence of abundant traces of the past that produces the familiar sensation that the past is determined. The absence of any analogous traces of the future produces the sensation that the future is open. The existence of traces serves to make it possible for our brain to dispose of extensive maps of past events. There is nothing analogous to this for future ones. This fact is at the origin of our sensation of being able to act freely in the world: choosing between different futures, even though we are unable to act upon the past.

The vast mechanisms of the brain about which we have no direct awareness ("I know not why I am so sad," mumbles Antonio at the beginning of *The Merchant of Venice*) have been designed during the course of evolution in order to make calculations about possible futures. This is what we call "deciding." Since they elaborate possible alternative futures that would follow if the present were exactly as it is except for some detail, we are naturally inclined to think in terms of "causes" that precede "effects": the cause of a future event is a past event such that the future event would not follow in a world that was exactly the same except for this cause.[101]

In our experience, the notion of cause is thus asymmetrical in time: cause precedes effect. When we recognize in particular that two events "have the same cause," we find this common cause[102] in the past, not in the future. If two waves of a tsunami arrive together at two neighboring islands, we think that there has been an event *in the past* that has caused both. We do not look for it in the future. But this does not happen because there is a magical force of "causality" going from the past to the future. It happens because the improbability of a correlation between two events requires something improbable, and it is *only* the low entropy of the past that provides such improbability. What else could? In other words, the existence of common causes in the past is nothing but a manifestation of low entropy in the past. In a state of thermal equilibrium, or in a purely mechanical system, there isn't a direction to time identified by causality.

The laws of elementary physics do not speak of "causes" but only of "regularities," and these are symmetrical with regard to past and future. Bertrand Russell noted this in a famous article, writing emphatically that "The law of causality . . . is a relic of a bygone age, surviving, like the monarchy, only because it is erroneously supposed to do no harm."[103] He exaggerates, of course, because the fact that there are no "causes" *at an elementary level* is not a sufficient reason to render obsolete the very notion of cause.[104] At an elementary level there are no cats either, but we do not for this reason cease to